

Sex-specific resilience of neocortex to food restriction

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Abstract

Mammals have evolved sex-specific adaptations to reduce energy usage in times of food scarcity. These adaptations are well described for peripheral tissue, though much less is known about how the energy-expensive brain adapts to food restriction, and how such adaptations differ across the sexes. Here, we examined how food restriction impacts energy usage and function in the primary visual cortex (V1) of adult male and female mice. Molecular analysis and RNA sequencing in V1 revealed that in males, but not in females, food restriction significantly modulated canonical, energy-regulating pathways, including pathways associated with AMP-activated protein kinase (AMPK), peroxisome proliferator-activated receptor alpha (PPAR α), mammalian target of rapamycin (mTOR), and oxidative phosphorylation. Moreover, we found that in contrast to males, food restriction in females did not significantly affect V1 ATP usage or visual coding precision (assessed by orientation selectivity). Decreased serum leptin is known to be necessary for triggering energy-saving changes in V1 during food restriction. Consistent with this, we found significantly decreased serum leptin in food-restricted males but no significant change in food-restricted females. Collectively, our findings demonstrate that cortical function and energy usage in female mice are more resilient to food restriction than in males. The neocortex, therefore, contributes to sex-specific, energy-saving adaptations in response to metabolic challenge.

eLife assessment

This study provides **important** findings based on **convincing** evidence demonstrating that females and males have different strategies to regulate energy consumption in the brain in the context of low energy intake. While food deprivation reduces energy consumption and visual processing performance in the visual cortex of males, the female cortex is unaffected, likely at the expense of other functions. This study is relevant for scientists interested in body metabolism and neuroscience.

Introduction

Mammals reduce their energy usage in times of food scarcity. These adaptations have been well documented for peripheral tissues and are sex-specific. For example, females, as compared to males, readily lose muscle and bone mass to reduce peripheral energy expenditure during food restriction, but lose less fat mass and overall bodyweight^{1–6}. Moreover, females are more likely to suppress energy-costly, reproductive functions during food restriction than males, evidenced by substantive reductions in uterine and ovarian mass, as well as a cessation of reproductive function and behaviour^{7–9}. Such sex-specific differences are thought to reflect the differential importance of different organs to sex-specific survival².

In contrast to peripheral tissue, whether and how the brain adapts its function and energy use in a sex-specific manner during food scarcity remains largely unknown. The brain requires substantial amounts of energy to process and encode information^{10–12}. Indeed, the human brain, which comprises 2% of our body's mass, consumes 20% of our caloric intake, over half of which is used by the cerebral cortex¹². Previously, we demonstrated that food restriction in male mice reduced energy usage in the primary visual cortex (V1) during visual processing, which was associated with a loss of orientation selectivity and visual coding precision¹³. These changes required a decrease in levels of serum leptin, a hormone secreted by adipocytes in proportion to fat mass¹⁴. Thus, in males the cortex, like peripheral tissue, has the capacity to reduce its energy usage and function in times of food scarcity.

By contrast, it remains unclear to what extent the brain similarly contributes to energy-saving adaptations during food scarcity in mammalian females. Understanding how food restriction impacts cortical function in males and females is not only of fundamental importance for understanding the sex-specific impact of diet on brain function, but is also critical for studies of cortical function, in which food restriction is used to behaviourally motivate animals^{15–17}.

Here, we examined how food restriction leading to a 15% loss in body weight over 2–3 weeks impacts energy usage and visual coding in the primary visual cortex (V1) in adult male and female mice. We found that leptin levels, which regulate energy-saving changes in¹³, were maintained in females during food restriction, despite being decreased by approximately threefold in males. Consistent with these results, and using a range of experimental techniques, we failed to find significant energy-saving changes in V1 during food restriction in female mice, in contrast to male mice. Specifically, molecular analysis and RNA sequencing of V1 revealed that food-restriction significantly modified canonical energy-regulating cellular pathways in males, but not in females, including pathways associated with AMP-activated protein kinase (AMPK), peroxisome proliferator-activated receptor alpha (PPAR α), and mammalian target of rapamycin (mTOR) signalling, as well as oxidative phosphorylation. Moreover, using *in vivo* ATP and calcium imaging, we found that food restriction significantly reduced V1 ATP use and orientation selectivity in males, but not in females.

Collectively, our study demonstrates that in times of food restriction, visual cortical function and energy usage are largely maintained in female mice, while they are reduced in males. The neocortex, therefore, contributes to sex-specific, energy-saving adaptations in response to metabolic stress. Our research highlights the importance of taking into account sex differences when studying the impact of dietary manipulations on brain function.

Results

We first examined sex-specific differences in how the neocortex adapts to food restriction in mice. Mice, 7-9 weeks of age, either had *ad libitum* (CTR) access to food or were food restricted (FR) to 85% of their free feeding bodyweight, for 2-3 weeks (**Figure 1A**). Prior to any of our recordings, tissue collection, and analysis, all animals were given *ad libitum* access to food until sated. This enabled us to examine the long-term impact of caloric restriction on neocortical function, as opposed to the short-term impacts of hunger.

Sex-specific impact of food restriction on weight loss and fat mass-regulated hormone leptin

Prior to food restriction, we found that both males and females consumed the same *ad libitum* daily food intake, despite females weighing approximately 80% of age-matched males (weight: male vs. female; $29.42 \pm 0.20\text{g}$ vs $24.23 \pm 0.54\text{g}$; t-test: $p < 0.001$; $n = 8$ male mice and 7 female mice). Critically, we found that to achieve a similar magnitude (15%) and rate of weight loss in both sexes we had to impose a 40% greater restriction of daily food intake in females (average of 39% of *ad libitum* intake; $n = 7$ female mice) compared to males (27% of *ad libitum* intake; $n = 8$ male mice) (**Figure 1B,C**). This is consistent with previous studies demonstrating that female mice are more resistant to weight loss than male. Moreover, for similar bodyweight loss (15%) maintained over 2-3 weeks, serum leptin levels, which reflect fat mass, were markedly and significantly decreased (-72%) in males (t-test: $p < 0.001$; $n = 17$ control male mice and $n = 19$ food-restricted male mice), but only modestly and non-significantly decreased (-28%) in females (t-test: $p = 0.32$; $n = 23$ control female mice and $n = 11$ food-restricted female mice) (**Figure 1D**). This is in keeping with previous findings that female mice are more resistant to fat loss than males and that weight loss arises from other tissues including those associated with reproductive functions (ovarian and uterine mass).

Cellular energy-regulating pathways in cortex are significantly impacted by food restriction in males but not in females

Decreased leptin levels are necessary for triggering energy-saving changes in the mouse visual cortex during food restriction. We therefore asked whether the absence of significant change in serum leptin in food-restricted females was associated with any change in V1 energy usage. We first assessed this at the molecular level by examining AMPK phosphorylation state. AMPK is a canonical cellular energy sensor activated by metabolic stress - such as food restriction, in part via decreased leptin signalling - through phosphorylation of threonine 172 (Thr¹⁷²). Its activation results in reduced ATP expenditure, for example, through the inhibition of mammalian target of rapamycin (mTOR) signalling, and regulation of mitochondrial oxidative phosphorylation. In V1, we found that AMPK Thr¹⁷² phosphorylation was markedly and significantly elevated (2.9 fold) with food restriction in males ($p < 0.05$; $n = 11$ control male mice and $n = 15$ food-restricted male mice), consistent with ATP savings, but only modestly increased (1.4 fold) in females; this increase was not statistically significant ($p = 0.11$; $n = 11$ control female mice and $n = 6$ food-restricted female mice) (**Figure 1E**). We also examined the activity of peroxisome proliferator-activated receptor alpha (PPAR α), which is downstream of leptin signalling and is known to regulate fatty acid metabolism and energy homeostasis. As with AMPK, we found that PPAR α was significantly regulated by food restriction selectively in males, but not in females.

To further investigate how energy-regulating cellular pathways are impacted by food restriction, we performed RNA sequencing on tissue dissected from V1. Differential gene expression analysis using DESeq2 revealed that food restriction significantly altered the

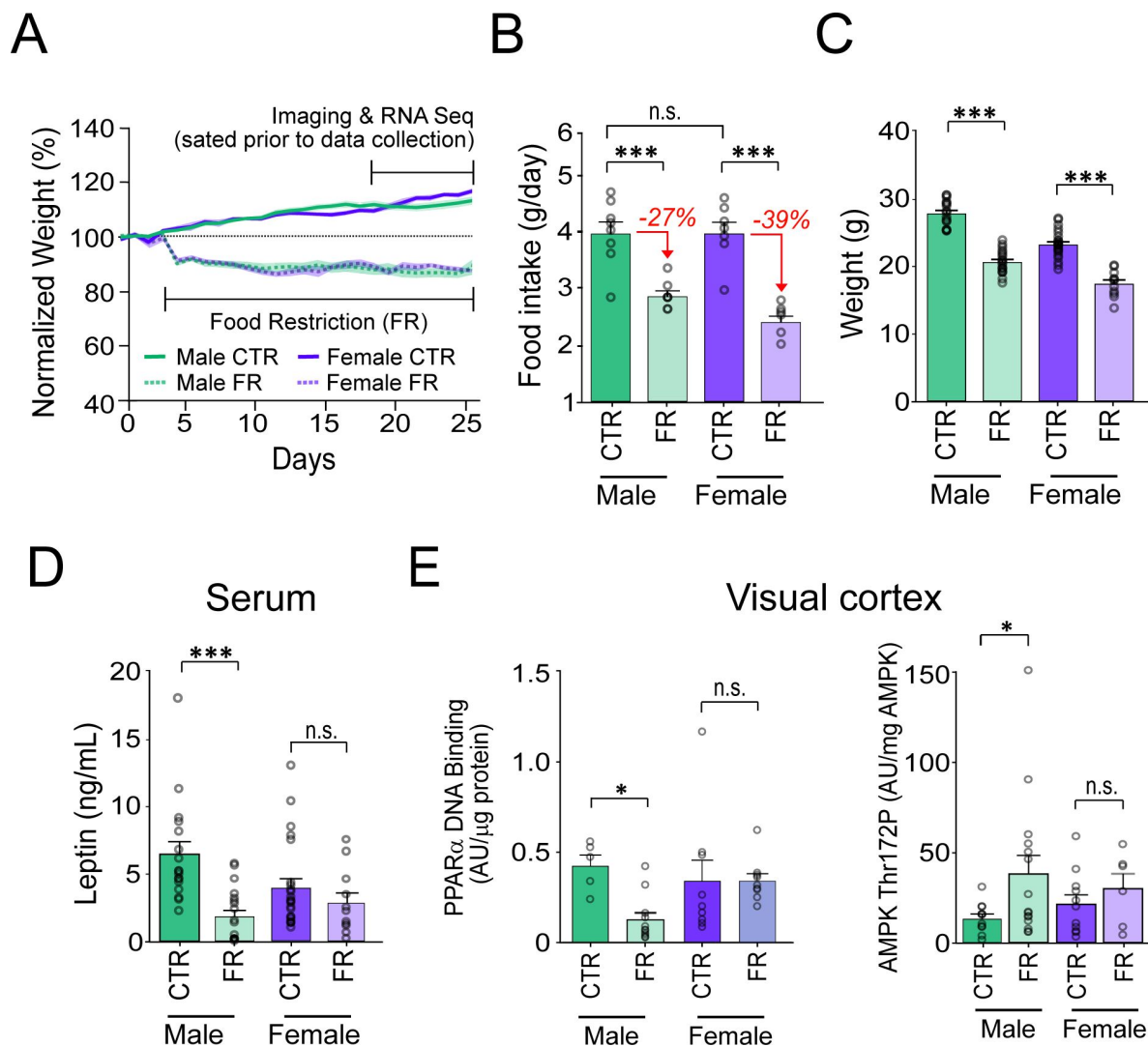


Figure 1.

Sex-specific impact of food restriction on weight loss and serum leptin.

(A) Animal weight across time. (B) Daily food intake (two-way ANOVA: CTR male vs. CTR female; $p = 0.99$; FR male vs. FR female; $p = 0.07$; $n = 8$ CTR males, 8 FR males, 7 CTR females, and 7 FR females). Percent reduction of food intake for food restriction is shown in red for each sex. (C) Animal weight (two-way ANOVA: $n = 17$ CTR males and 19 FR males; 22 CTR females and 11 FR females). (D) Serum leptin levels (two-way ANOVA: CTR females vs. FR females; $p = 0.32$; $n = 17$ CTR males and 19 FR males; 22 CTR females and 11 FR females). (E) Left. AMPK Thr172 phosphorylation, normalized by total AMPK, in primary visual cortex (V1) tissue (two-way ANOVA: CTR female vs. FR female; $p = 0.11$; $n = 11$ CTR males, 15 FR males, 11 CTR females, and 6 FR females). Right. PPAR α activity in V1 tissue, as assessed by levels of DNA binding, normalized to protein level (two-way ANOVA: CTR female vs. FR female; $p = 0.99$; $n = 5$ CTR males, 11 FR males, $n = 9$ CTR females, and 9 FR females). * $p < 0.05$; n.s. = not significant. Error bars are S.E.M.

expression of 657 and 585 targets in males and females, respectively (at a threshold of $p\text{-adj} \leq 0.1$; only 125 of these were common to both sexes (**Figure 2B**). To assess the functional relevance of the transcriptomic changes induced by food restriction, we performed Gene Set Enrichment Analysis (GSEA) to identify pathways enriched in the differentially expressed populations. We found that food restriction resulted in a significant alteration of 34 and 14 gene sets (at $p\text{-adj} \leq 0.05$) in males and females respectively; 13 of these were common to both sexes (**Figure 2C**; Table 1; Table 2). In male mice, we specifically found that food restriction significantly regulated gene sets associated with canonical energy-regulating pathways (**Figure 2D**), including those associated with oxidative phosphorylation and mTOR signalling (mTORC1 signalling and PI3K/AKT/mTOR signalling), as well as fatty acid metabolism; these pathways are known to be regulated by leptin and AMPK signaling^{20,27–29} (**Figure 2D**). Notably, these pathways were not significantly regulated by food-restriction in females. In addition to these findings, we found that gene sets associated to estrogen-regulating pathways were significantly regulated in food-restricted males and not in females (**Figure 2C**). This is consistent with previous findings showing that estrogen, like leptin, suppresses food intake and promotes energy expenditure³⁰. Finally, consistent with past studies in cortex, we found that food restriction in males altered gene expression associated with inflammation and reactive oxidative species (**Figure 2D**)^{31,32}.

Collectively these findings reveal that food restriction strongly affects energy-regulating cellular pathways in the cortex of male, as opposed to female, mice.

ATP usage in V1 is significantly decreased under food restriction in males but not in females

We next examined energy usage in visual cortex during visual processing using two photon ATP imaging in V1 of awake, head-fixed Thy1-ATeam1.03^{YEMK} mice, which express a FRET-based ATP sensor^{33,34} (**Figure 3A**). In the presence of ATP synthesis inhibitors, focally applied over the visual cortex, ATP usage can be seen as a decay of the FRET signal during visual stimulation with natural stimuli¹³ (**Figure 3A**). Data obtained in females were compared to our previously published datasets obtained under similar conditions in male mice¹³. As previously shown¹³, food restriction robustly decreased the rate of cortical ATP usage in males (24% decrease) (t-test: $p < 0.01$; $n = 11$ control male mice and $n = 10$ food-restricted male mice); a more modest trend was observed in females (12% decrease), which was not significant compared to controls (t-test: $p = 0.18$; $n = 12$ control female mice and $n = 8$ food-restricted female mice) (**Figure 3B–C**). These findings demonstrate that food-restriction significantly impacts V1 energy usage in males but not in females.

Orientation selectivity in V1 is significantly reduced under food restriction in males but not in females

We next examined how food restriction impacts visual cortical function. In male mice, reductions in energy usage in V1 during food restriction are associated with a decreased visual coding precision as evidenced by a reduction of orientation selectivity¹³. Given the non-significant impact of food-restriction on energy usage in female mice, we reasoned that orientation selectivity would also be less impacted. We analysed two-photon calcium imaging data of V1 GCaMP6s-labeled layer 2/3 neurons in female mice viewing drifting gratings (**Figure 3D–F**; $n = 7$ control male, $n = 8$ food-restricted male, $n = 7$ control female, and $n = 9$ food-restricted female mice). Data obtained in females were compared to our previously published datasets obtained under similar conditions in male mice¹³. Consistent with our hypothesis, in contrast to males in which orientation selectivity was significantly reduced by 27% ($p < 0.001$), orientation selectivity was modestly and not significantly reduced by food-restriction in females (13%; $p = 0.08$) (**Figure 3E–F**).

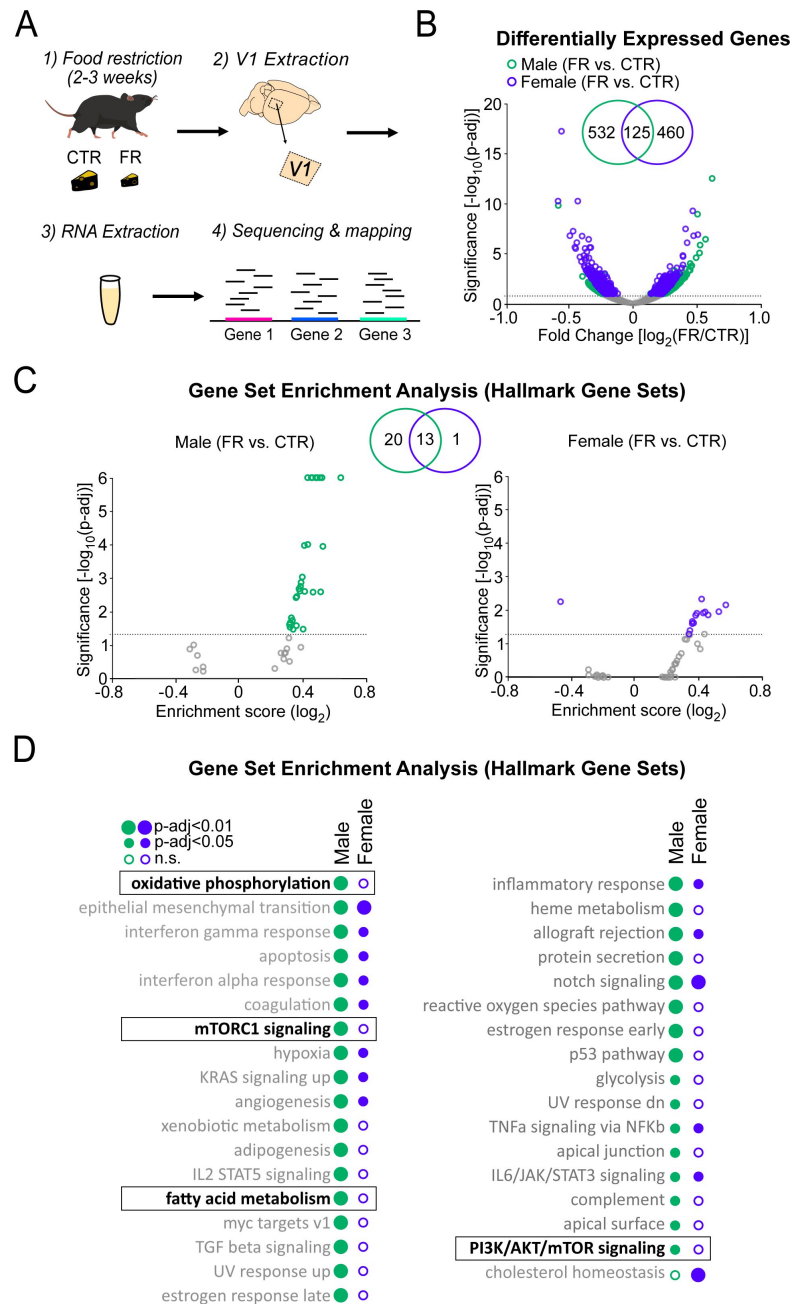


Figure 2.

Cellular energy-regulating pathways are more robustly impacted by food restriction in males than in females.

(A) Schema of RNA sequencing of V1 tissue. (B) Volcano plot of fold change expression with food restriction vs. significance for all analyzed genes; broken horizontal line denotes adjusted $p \leq 0.1$ significance level, below which non-significantly regulated expressed genes are marked by grey. Inset: Venn diagram showing the number of significantly differentially expressed genes with food restriction in males and females. (C) Gene Set Enrichment Analysis using Hallmark Gene Sets, depicted using a volcano plot of Enrichment Score vs. significance for males (left) and females (right); broken horizontal lines denotes adjusted $p \leq 0.05$ significance level. Inset: Venn diagram showing the number of significantly regulated gene sets with food restriction in males and females. (D) Complete list of significantly enriched gene sets by food-restriction in males and females for analysis in C. Gene sets of interest are denoted by a box. All data from 4 CTR males, 4 FR males, 4 CTR females, and 4 FR females.

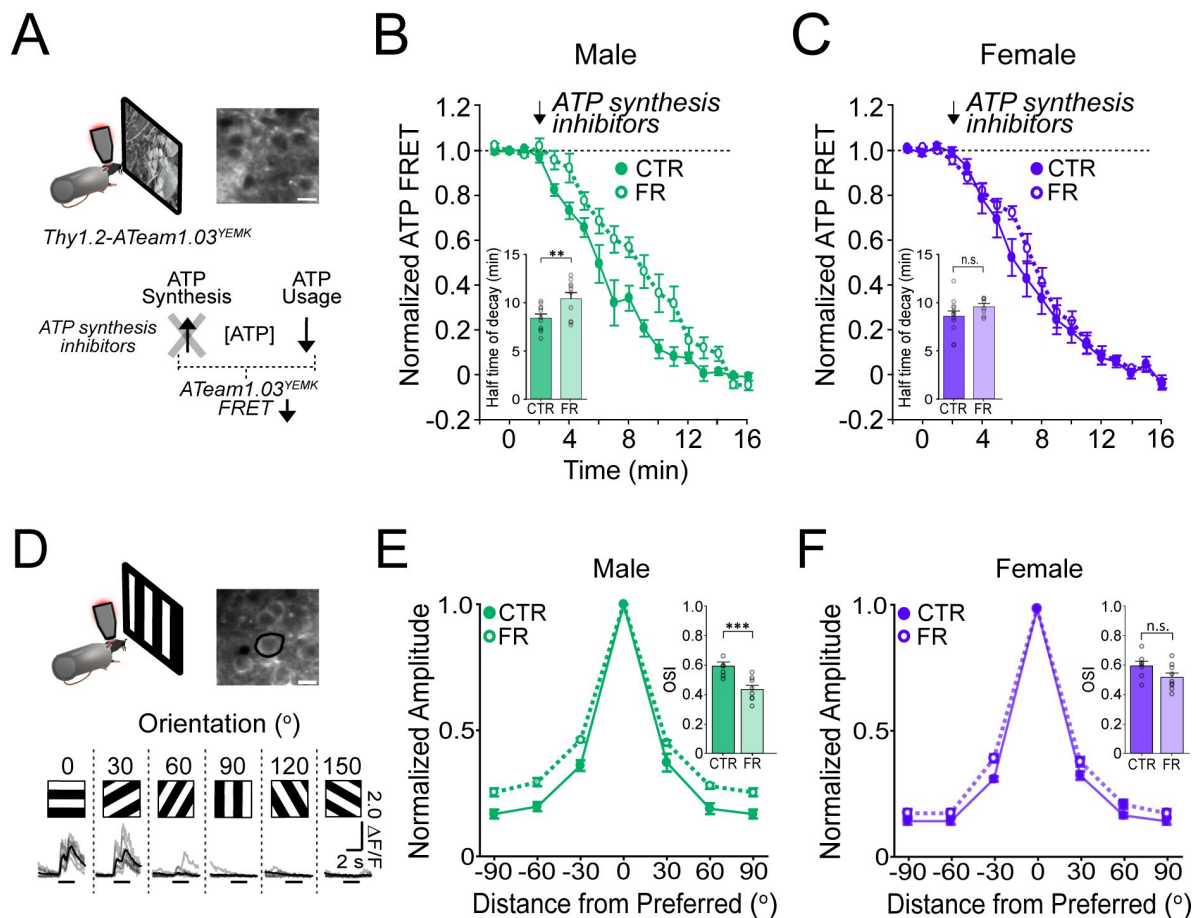


Figure 3.

ATP use and orientation selectivity in V1 is more robustly decreased by food restriction in males.

(A) Schemata of ATP imaging experiment. Top: example field of view of V1 layer 2/3 neurons in the ATeam1.03^{YEMK} transgenic mouse (scale bar: 10 μ m). Bottom: ATP synthesis inhibitors were used to isolate ATP usage, recorded as a decrease in FRET signal during presentation of natural, outdoor scenes. (B) Normalized ATeam1.03^{YEMK} FRET signal in males and during presentation of natural, outdoor scenes. ATP synthesis inhibitors (arrow) were added to isolate ATP usage. Inset: Time to 50% decay of the ATeam1.03^{YEMK} FRET signal (two-way ANOVA; $n = 11$ CTR males and 10 FR males). (C) As in (B) but for females (inset: two-way ANOVA; $p = 0.18$; $n = 12$ CTR females and $n = 8$ FR females). (D) Top: Schema of two-photon imaging and sample field of view of V1 layer 2/3 neurons expressing GCaMP6s (scale bar: 10 μ m). Bottom: Sample fluorescent signals (grey) from a selected neuron (black circle) in response to 6 drifting gratings of varying orientations. Trial averages are in black. Horizontal bar denotes 2s grating presentation. (E) Mean orientation tuning normalized to the response to the preferred orientation for CTR and FR males (data from 13). Note that -90° and $+90^\circ$ conditions correspond to the same visual stimulus. Inset: Orientation selectivity index (OSI; two-way ANOVA; $n = 8$ CTR and 8 FR males). (F) As in (E) but for females (Inset: two-way ANOVA; $p = 0.08$; $n = 7$ CTR females and 9 FR females). ** $p < 0.01$; *** $p < 0.001$; n.s. = not significant. Error bars are S.E.M. Data from male mice are from a previously published dataset obtained under similar conditions¹³.

Altogether, our results show that energy usage and coding precision in the visual cortex are largely maintained in female mice under food restriction while they are robustly reduced in males.

Bayes factor analysis reveals statistically robust impact of metabolic stress on V1 energy use and visual coding in males, but not females

We used Bayes factor hypothesis testing to better assess the statistical impact of food restriction on visual cortical function and energy use in male and female mice (**Supplemental Figure 1**).

Bayes factor analysis uses a Bayesian framework to quantify how much more likely the data is under the alternative hypothesis (*i.e.* there is an effect of food restriction) versus the null hypothesis (*i.e.* there is no effect of food restriction), quantified by the Bayes factor (BF_{10})^{35,36}. A $BF_{10} > 3$, for example, means that the data is >3 times more likely under the alternative hypothesis than the null hypothesis. Critically, the BF_{10} allows for a more informative comparison of the effects of food restriction between sexes, as opposed to comparing p-values, which can be misleading^{35,36}. Across the parameters of energy use and visual coding that we tested (**Figures 1–3**), BF_{10} for males was consistently several-fold greater than that for females. Thus, visual cortical function and energy use was more robustly modulated by food restriction in males than in females.

Our findings however cannot rule out a potential, smaller impact of food restriction on cortical function and energy use in females, below the threshold for statistical detection in this study. The Bayes factor analysis helps to differentiate the absence of evidence from the evidence of absence. In particular, moderate evidence in support for the absence of an effect requires a $BF_{10} < 0.33$ (*i.e.* the data is 3 times more likely under the null hypothesis than the alternative hypothesis)^{35,36}. For females, we found only 3 out of the 10 parameters had $BF_{10} < 0.33$ (**Supplemental Figure 1**), suggesting that food restriction may have a modest impact on V1 energy use or function, but below the limits of statistical detection in our study. Altogether, our results show that energy usage and coding precision in the visual cortex are largely maintained in female mice under food restriction while they are robustly reduced in males.

Discussion

Here, we find that in female mice, energy usage and visual coding in V1 is largely unchanged under food restriction. Using molecular analysis and RNA sequencing, we found that energy-regulating pathways were not significantly changed with food restriction in females, in contrast to males; these included pathways associated with AMPK, mTOR, and PPAR α signalling, as well as oxidative phosphorylation. Consistent with this, we found that food restriction in females did not impact visual cortical ATP usage nor orientation selectivity, both of which were reduced by food restriction in males. Our findings suggest that the contribution of neocortex to energy-saving adaptations during food restriction is sex-selective and more pronounced in males; females may rely on other energy-saving strategies, such as reducing reproductive function, to cope with food restriction.

Limitations of the study

Our study focused on the effects of a moderate food restriction protocol over 2–3 weeks on neocortical function and energy use. This protocol is widely used in metabolic studies investigating the impact of caloric restriction on peripheral physiology, as well as in neuroscience studies to motivate rodents to perform specific behaviors^{6,15–17}. However, it remains unclear to what extent our findings generalize to other protocols of food restriction. Indeed, the duration, magnitude, and pattern of food restriction, as well as macronutrient composition are

known to impact the physiological responses to dietary manipulations^{22,37,39}. Moreover, whilst we focussed our study on the adult visual cortex, it remains unclear if the brain shows similar sex-specific energy-saving adaptations in other cortical areas, and how such adaptations change with age^{6,40}.

A key strength of our study is that we consistently found, across a number of experimental methods, that changes in cortical function and energy use were significant in males, but not in females. However, a critical limitation of our study is that we did not have the statistical power to directly compare the effect size of food restriction between the sexes (e.g. ANOVA interaction effects); to do so would require a substantial increase in experimental group sizes by several fold. We addressed this limitation, in part, using Bayes factor analysis, which revealed that the impact of food restriction in males was indeed more robust than in females. This analysis also revealed that our study did not have sufficiently strong evidence to definitively rule out a smaller effect of food restriction in females, which may not have been statistically detectable.

Role of leptin and sex hormones in food restriction impact on cortical activity

Consistent with past studies, we found that food-restricted female mice resisted weight loss, and therefore required a greater restriction of food to achieve similar weight loss (relative to baseline) as males. In addition, for the same relative weight loss as males, females exhibited minimal reductions in serum leptin levels (secreted by adipocytes), consistent with past studies showing that females resist fat loss during food restriction^{1,3,5,6}. Similar effects have been observed across several mammalian species, including humans, and is thought to reflect the differential importance of fat stores for reproductive success in males and females^{1,5}. Notably, we previously demonstrated that reductions in leptin levels during food restriction are necessary for driving energy-saving changes in visual cortical coding in male mice; leptin therefore acts as a critical signal linking the neocortex to fat stores¹³. That leptin levels are not significantly affected by food restriction in females may explain the minimal impact food restriction has on cortical energy usage and coding precision in female mice.

In addition to leptin, sex hormones likely contribute to sex-specific adaptations to food restriction⁴¹. As with leptin, estrogen suppresses food intake and promotes energy expenditure³⁰, in part by augmenting leptin signalling by increasing leptin-induced phosphorylation of STAT3⁴². Moreover, estrogen can enhance ATP production by sensitizing insulin signalling, enhancing glycolysis and oxidative phosphorylation, and along with progesterone, upregulating mitochondrial gene expression and function^{30,43,44}. Interestingly, in our study, RNA sequencing revealed that estrogen signalling pathways were significantly regulated in male visual cortex with food-restriction, and were unchanged in females. Estrogen may thus have overlapping effects with those induced by food restriction, thereby diminishing further responses to food restriction in females⁶. Notably, it was recently shown that sex differences in calorie-restriction's metabolic effects that are observed in young adult mice, are largely absent in older mice, when females' estrogen levels have declined⁶. Further studies will be needed to explore whether the resilience of cortical function found in this study in young cycling adult females is also observed in non-cycling mice with reduced oestrogen levels, such as in aged mice.

Sex-specific regulation of cellular energy-regulating pathways

AMPK and mTOR signalling pathways are canonical regulators of cellular energy usage^{20,23}. We found these pathways, along with cortical ATP usage, to be significantly modulated by food-restriction in male, but not female mice. AMPK is activated during metabolic stress, and saves energy by regulating oxidative phosphorylation^{24,25}, and inhibiting cellular pathways associated with energy usage, such as mTOR signalling²⁰. mTOR signalling, by upregulating protein synthesis, increases ATP use by promoting neuronal excitability, neurite outgrowth, ion

channel expression, and synaptic plasticity²³. mTOR signalling is therefore ideally situated to regulate how much ATP is used in neuronal function; its inhibition may underlie ATP savings and the associated loss of coding precision with food-restriction in male mice. Notably, previous studies have found sex- and tissue-dependent regulation of AMPK and mTOR signalling, including in response to food restriction⁴¹. For example, in skeletal muscle, mTOR signalling is enhanced by fasting selectively in male, but not female mice^{41,45}. Moreover, metabolic challenge by endurance training induces greater AMPK activity in cardiac tissue in male, as opposed to female mice⁴⁶. Our study extends the sex-specific regulation of these energy-regulating pathways to the neocortex.

Importance of biological sex on the impact of food restriction on cortical function

Food restriction protocols are widely used to motivate animals in behavioural studies examining cortical function^{15–17}. Our findings suggest that these protocols are likely to have greater impacts on basal neocortical function in males, as compared to females. Consistent with this, a recent study has found that food restriction in juvenile mice strongly impacts cognitive flexibility in learning and decision-making in adulthood, specifically for males but not females^{47,48}. Such sex-specific impacts of food restriction on neocortical function are important to take into consideration in the design and interpretation of such studies.

Diet is known to have a considerable impact on neurological diseases, including epilepsy, Alzheimer's disease, and other neurodegenerative diseases^{49,50}. Whilst calorie restricted diets have proven to be of benefit in this regard, the sex-dependency of such interventions have not been thoroughly investigated, though on the basis of our findings it is conceivable that males and females may respond differently to dietary interventions. The inclusion of both males and females in biomedical research, and the study of sex differences, is therefore of considerable importance for ensuring efficacious treatments are developed for both sexes.

In conclusion, we find that visual cortical coding and energy usage in females are resistant to food restriction, in contrast to males. Our findings establish that sex-specific adaptations in energy usage, previously described in peripheral tissue under metabolic challenge, extends to the neocortex.

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Author contributions

Z.P. and N.R. designed the experiments. Z.P. performed and analysed calcium imaging experiments and ELISAs. D.K. performed and analysed calcium and ATP imaging experiments. Z.P. and M.R. and E.O. processed tissue for RNA sequencing and analyzed data. P.M. performed and analyzed ELISAs. N.R. supervised the project. Z.P. and N.R. wrote the manuscript with input from all authors.

Declaration of Interests

The authors declare no competing interests.

Inclusion and Diversity

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in science. One or more of the authors of this paper self-identifies as a member of the LGBTQ+ community. While citing references scientifically relevant for this work, we also actively worked to promote gender balance in our reference list.

Figure Legends

Star methods

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact Nathalie L. Rochefort (n.rochefort@ed.ac.uk).

Materials availability

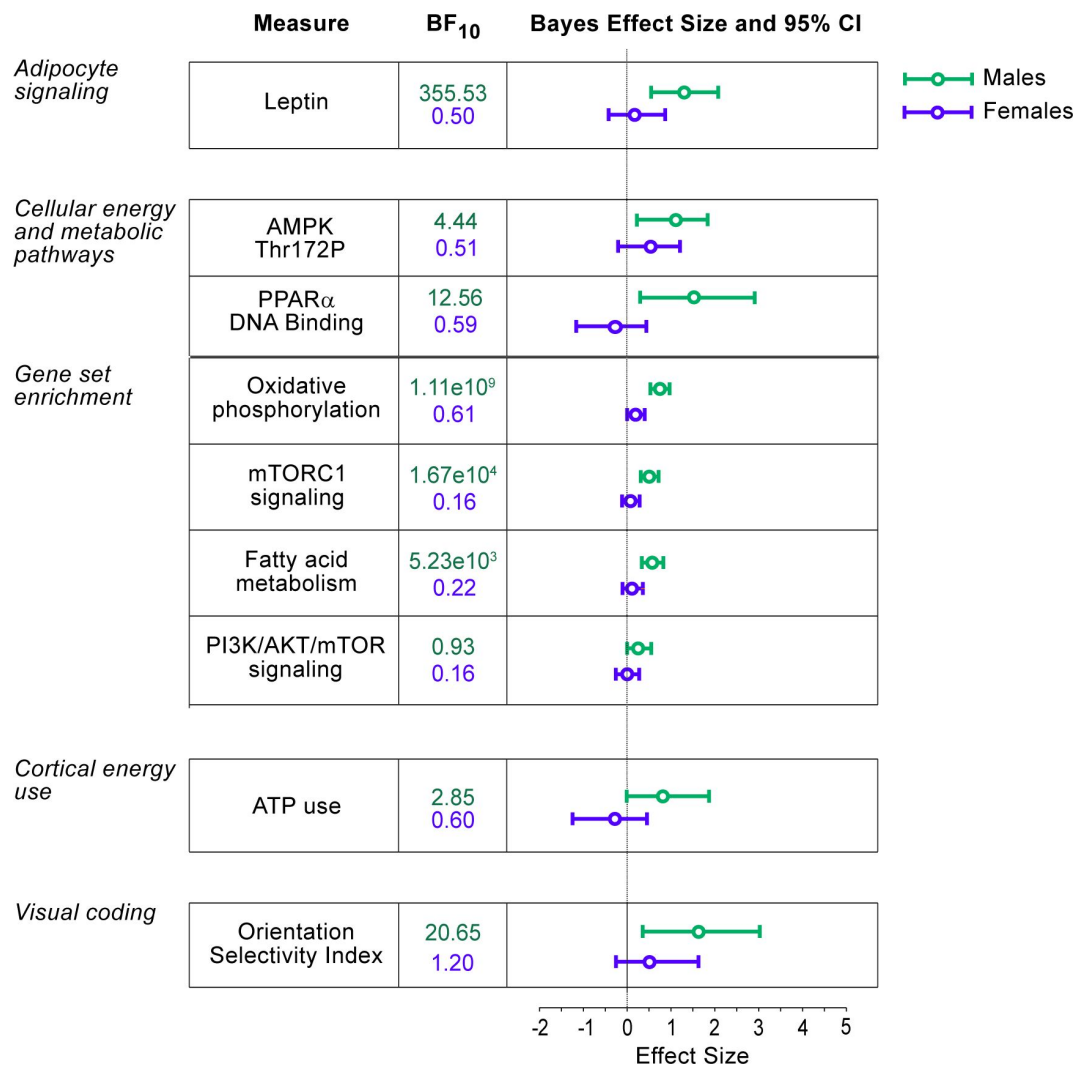
This study did not generate new unique reagents.

Data and code availability

1. GEO accession number for RNA-seq datasets: GSE233435
2. Processed data will be made available at <https://datashare.ed.ac.uk/handle/10283/3871>. Requests for raw data should be made to and will be fulfilled by the lead contact (n.rochefort@ed.ac.uk).
3. MATLAB scripts to analyse data have been previously published¹³ and are available at <https://zenodo.org/record/5561795#.Ymj3AtrMJPY> and <https://github.com/rochefort-lab/Padamsey-et-al-Neuron-2022>.
4. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Experimental model and subject details

Experiments were approved by the University of Edinburgh's Animal Welfare and Ethical Review Board (AWERB) and carried out under Home Office (UK) approved project and personal licenses. All experiments conformed to the UK Animals (Scientific Procedures) Act 1986 and the European



Supplemental Figure 1.

Bayes factor analysis reveals statistically robust impact of metabolic stress on V1 energy use and visual coding in males, but not in females.

Bayes factor (BF₁₀) along with quantified effect size and 95% confidence intervals (CI) are shown for males (green) and females (purple). Across parameters, BF₁₀ and effect sizes are greater for males than females, meaning that food restriction induces a more robust effect on visual cortical function and energy use in males than females.

Directive 86/609/EEC and 2010/63/EU on the protection of animals used for experimental purposes.

This study used male and female C57BL/6J mice (RRID:IMSR_JAX:000664; Jackson Laboratory) and male B6-Tg (Thy1.2-A_{Team1.03}^{YEMK})^{AJhi} transgenic mice, that were bred on a C57BL/6J background (RRID:MGI:5882597; <https://scicrunch.org/resources>). Animals were group housed (2-5 animals/cage) and maintained on a reverse light/dark (12h/12h) cycle in a room kept at 21±2°C and 55±10% humidity.

For data associated with *in vivo* recordings (Figure 3), data obtained in female mice were compared to our previously published datasets obtained under similar conditions in male mice¹³.

Method details

Food restriction

Mice (7-9 weeks of age) either had *ad libitum* access to food (RM1 expanded pellets; DBM Scotland UK) or were food restricted for 2-3 weeks to 85% of their baseline bodyweight prior to experimentation as previously described¹³. Briefly, for food restriction, one ration of food was given 4-8 hours prior to the end of the dark cycle, with the amount of food progressively reduced to achieve target weight. 1-2 hour prior to experimentation, all animals had *ad libitum* access to food until sated.

AAV injection and cranial window

For calcium imaging experiments, C57BL6 were anesthetized using isoflurane, and given pre-operative analgesia (vetergesic: 0.1 mg/kg; carprieve: 5 mg/kg; rapidexon: 2 mg/kg) along with warm Ringer's solution (25ml/kg) subcutaneously; vetergesic jelly was additionally given orally post-operatively 24 hours later. Opaque eye cream was applied to the eyes (Bepanthen, Bayer, Germany). A craniotomy (2×2 mm) was applied over left V1 (centre at 2.5 mm mediolateral and 0.5 mm anterior to lambda). Flexed GCaMP6s (AAV1.Syn.Flex.GCaMP6s.WPRE.SV40; Addgene; RRID:Addgene_105558-AAV1) diluted 1:10 in saline, along with CaMKII-dependent Cre-recombinase (AAV1.CaMKII 0.4.Cre.SV40; Addgene; RRID:Addgene_105558-AAV1) diluted 1:100 saline was injected at the centre of the craniotomy via a sharp glass pipette. Injections were targeted to layer 2/3, with 100 nL of virus solution injected (2.5nL/30s) at each of 3 depths (150, 250 and 350 µm) via a sharp glass pipette using a Nanoject III (Drummond Scientific). A glass window was then used to seal the craniotomy, and superglued in place. The window was made of 1 or 2 glass coverslips (Menzel-Glaser # 0), in the latter case the 2 cover slips were glued together with optically-clear, UV-cured glue (Norland Optical Adhesive no. 60); the inner window was 2.0 mm x 2.0 mm, the outer window was 2.5 mm x 2.5 mm. A metal headplate was then fixed to the skull with super glue and dental cement (Paladur, Heraeus Kuzler). It took 3-6 weeks for viral expression prior to experimentation.

For ATP imaging experiments, we used A_{Team1.03}^{YEMK} mice (RRID:MGI:5882597; <https://scicrunch.org/resources>) that express a FRET-based sensor under the Thy1 promoter³³. One the day of experimentation, animals were anesthetized with isoflurane and given pre-operative analgesia (Carprieve: 5 mg/kg) and 25 mL/kg of warm Ringer's solution subcutaneously. A headplate was first fixed to the skull with superglue and dental cement, after which a small craniotomy (~0.5 x ~0.5 mm) was made over left V1 (centre at 2.5 mm mediolateral and 0.5 mm anterior to lambda), which was covered with agarose (1-2%) and silicone. Following recovery from anaesthesia (30-60 minutes) the animal was head-fixed; the agarose and silicone were removed and replaced with HEPES-buffered ACSF (in mM: 124 NaCl, 20 Glucose, 10 HEPES, 2.5 KCl, 1.2 NaH₂PO₄, 2 CaCl₂, and 1 CaCl₂; pH 7.2-7.4).

Habituation and Head-fixation

During imaging, mice were placed in a cardboard tube and head fixed. Prior to the imaging day, mice were handled daily for 2-3 weeks prior, and exposed to cardboard tubes, similar to the one on the experimental rig. They were also habituated to the rig via once daily, 10-15 minute sessions for 2 days prior to recording. For ATP imaging, mice were habituated to the experimental setup without head fixation (10-15 min session/day for 1-2 days).

In vivo two-photon imaging

ATP and calcium imaging were using a two-photon resonant scanning microscope, as previously described^{51,53}, which was equipped with a Ti:Sapphire laser (Chameleon Vision-S, Coherent, CA) and GaAsP photomultiplier tubes (Scientifica), and controlled with LabView (v8.2; National Instruments, UK). Time-series xy images were acquired at 40Hz using a 25x water-immersion objective (Nikon; CF175 Apo 25XC W; 1.1 NA) at depths between 160 and 280 μm below the pia. For GCaMP6s imaging, excitation was tuned to 920nm. For ATP imaging (CFP/YFP FRET sensor), the laser was tuned to 850 nm for CFP excitation. CFP and YFP fluorescence were recorded simultaneously using a 515nm long-pass dichroic mirror with the following emission filters: 485/70nm for CFP and 535/45nm for YFP (mirror and filter set: T515lpxr C156624; Scientifica). Imaging data was acquired during the presentation of visual stimuli, with at least 10 replicate trials for each visual stimulus. For ATP imaging, three trials were taken at baseline, then ACSF containing ATP synthesis inhibitors (1 mM oligomycin and 20 mM sodium iodoacetate) was applied over the open craniotomy isolate ATP usage. Thirty trials were successively performed immediately after drug application. For *in vivo* recordings, data obtained in females were compared to our previously published datasets obtained under similar conditions in male mice¹³.

Visual Stimuli

MATLAB (Mathworks; RRID:SCR_001622; Psychophysics Toolbox; RRID:SCR_002881) was used to generate drifting grating stimuli. These were displayed on an LCD monitor (51 x 29 cm; Dell) placed 20 cm from the right eye, contralateral to the hemisphere with the cranial window. For food restriction experiments, on a given trial, 12 full-field drifting gratings (drift angle: 0, 30, 60, 90, 120, 180, 210, 240, 270, 300, 330°; temporal frequency = 1 Hz), 2 s in duration, were presented in random order, interspersed with grey screens (4 s duration). Gratings within a trial were randomly assigned one common spatial frequency (0.02, 0.04, 0.16, or 0.32 cpd). For natural stimuli, a 60s movie was presented of an outdoor movie of movement through a field and vegetation.

Serum leptin measurements

Serum leptin levels were analyzed using ELISA kits (Mouse Leptin Quantikine Elisa Kit, R&D Systems, USA) according to manufacturer's instruction. For serum collection, trunk blood was collected from mice that were briefly anesthetized (isoflurane) prior to decapitation. Blood was allowed to clot at room temperature for 1-2 hours and then centrifuged for 20 minutes at 2000 x g. The serum was drawn off and stored at -80°C until used.

AMPK and PPAR α measurements

Levels of total AMPK, phosphorylated AMPK (Thr172), and nuclear PPAR α in V1 were measured using ELISA kits according to manufacturer's instructions (total AMPK: ab181422, Abcam, UK; phosphorylated AMPK: KHO0651, ThermoFisher, UK; PPAR α : ab133107, Abcam, UK). For tissue collection, mice were briefly anesthetized (isoflurane) and decapitated. The visual cortex was dissected out, snap frozen using dry ice, and stored at -80°C until used.

RNA sequencing

Bulk RNA sequencing was performed on V1. V1 tissue was collected and snap frozen from mice that were briefly anesthetized (isoflurane) and decapitated. Tissue was homogenized using a bead mill (Bead Mill 24, Fisher Scientific) and RNA was isolated using an RNeasy kit (Qiagen) according to manufacturer's instructions. RNA integrity was confirmed (RIN > 7) and mRNA was sequenced at the Oxford Genomics Centre using Illumina NovaSeq 6000.

Quantification and statistical analysis

Ca²⁺ Imaging Analysis

Image analysis for calcium imaging was performed as previously described^{45,47,49}. For motion correct, a discrete Fourier 2D-based image alignment was used (SIMA 1.3.2)⁵³. Regions of interest (ROI) were manually drawn around imaged neuronal soma using ImageJ software (NIH public domain; RRID:SCR_002285). Pixel fluorescence was then averaged within each ROI to generate a time series. For each ROI, baseline fluorescence (F_0) was calculated by taking the 5th percentile of the smoothed time series (1 Hz lowpass, zero-phase, 60th-order FIR filter). $\Delta F/F$ was then calculated as $(F - F_0)/F_0$. FISSA was used to decontaminate the neuropil⁵⁵. Subsequent analyses were performed using custom scripts in MATLAB (MathWorks)¹³.

Drifting grating analysis

The visual response to drifting gratings was defined as the highest mean $\Delta F/F$, averaged within a 2s window, that occurred within a 4 s window (comprising the 2s grating presentation plus the 2 s of the grey screen presentation which immediately followed the grating presentation). The visual response was subtracted by baseline $\Delta F/F$, defined as the mean value within a 1 s window prior to the visual response.

Responses to gratings of the same angle, but different drift directions were averaged together. Given that multiple spatial frequencies were used, only the spatial frequency with the largest response, meaned across all orientations, was selected for each neuron for subsequent analyses. A neuron's preferred orientation was that which evoked the largest mean response.

Grating responsive neurons were defined as those for which grating responses were better fit with a double Gaussian curve (direction responses) than with a flat line at zero (null model), which was determined using Bayesian Information Criterion (BIC)¹³. The double Gaussian curve was defined as:

$$R(\theta) = C + R_p e^{\frac{-\text{ang}_{dir}(\theta - \theta_{pref})^2}{2\sigma^2}} + R_n e^{\frac{-\text{ang}_{dir}(\theta - \theta_{pref})^2}{2\sigma^2}}$$

Where $R(\theta)$ is the response at a given direction angle θ , C is an offset, R_p is the response to the preferred direction after subtracting the offset, θ_{pref} is the angle of the preferred direction, R_n is the response to the null direction after subtracting the offset, and $\text{ang}_{dir}(x) = \min(x, x - 360, x + 360)$, which constrains angular differences to 0-180°, and σ is the standard deviation of the curve.

BIC was given by:

$$BIC = n \ln(\overline{\sigma^2}) + k \ln(n)$$

Where n is the number of responses, $\overline{\sigma^2}$ is the mean residual sum of squares of the model, and k is the number of free parameters used by the model, which was zero in the case of the null model. A neuron was considered significantly grating-responsive if the $BIC_{\text{null}} - BIC_{\text{Gaussian}} \geq 10$, which

provides strong evidence against the null model⁵⁴.

Orientation selectivity

The orientation selective index (OSI) of grating-responsive neurons was calculated as $1 - \text{circular variance}$ based on the mean response of the neuron to each; values less than zero were set to zero. OSI was given by:

$$OSI = \text{abs} \left(\frac{\sum_k R(\theta_k) e^{2i\theta_k}}{\sum_k R(\theta_k)} \right)$$

Where $R(\theta_k)$ is the response to the k_{th} angle θ . The average OSI for an animal was taken as the median OSI value calculated across neurons.

ATP measurements

For ATeam1.03^{YEMK} imaging, CFP and YFP fluorescence was averaged across ROIs drawn around visible soma in the field of view, after background subtraction. FRET was calculated as a ratio of YFP/CFP fluorescence, and decayed as a function of time with visual stimulation in the presence of ATP synthesis inhibitors. The FRET signal was bound between 0 and 1 by first subtracting the mean FRET signal during the last 3 imaging trials (the FRET signal plateaued at this point), and then dividing by the mean FRET signal at baseline. The FRET decay half time was defined as the time required for the FRET to decay to 50% of its baseline value.

Differential gene expression analysis

RNA sequencing reads were mapped, using STAR 2.4.oi, to the *Mus musculus* primary assembly (Ensembl v80)⁵⁵. FeatureCounts 1.4.6-p2 was used to count reads uniquely aligned to annotated genes⁵⁶. Differential gene expression analysis was then conducted using DESeq2 1.12.4 (betaPrior = False) to normalize counts⁶⁰. For each gene, a fold change in gene expression was defined as $\log_2 \frac{\text{Normalized count under food restriction (FR)}}{\text{Normalized count under control conditions (CTR)}}$ for males and females. Gene set enrichment analysis (GSEA) was conducted using GSEA software (<https://www.gsea-msigdb.org/gsea/index.jsp>) using the Hallmarks Gene Sets (v7.5.1).

Statistics

Statistical tests, p values, and the number and definition of independent units are stated in the figure legends. For all measures, the number of animals was taken as the statically independent unit. Sample size was determined by *a priori* based on calculations, with an aim of achieving 80% power on the basis of a 20% group difference and a significance threshold of $p < 0.05$. Group means and variances were taken from pilot data or previously published data¹³. Where applicable, ANOVA was used to assess significance, followed by *post-hoc* Fisher LSD tests to prevent loss of power. Statistical tests were carried out in Prism 6 (GraphPad Prism; RRID:SCR_002798). Averages denoted in figures represent means, with error bars representing the standard error of the means (S.E.M.). We complemented our statistical analyses using Bayes factor hypothesis testing (**Supplemental Figure 1**) as described in^{35,36} using JASP software to calculates Bayes factor, effect size, and confidence intervals (<https://jasp-stats.org/>). As recommended, we used a default, Cauchy distribution with a spread of $1/\sqrt{2}$ as a prior distribution of effect size. A $BF_{10} < 0.33$ is considered to be moderate evidence for the absence of an effect^{35,36}.

Declaration of interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

Padamsey et al. followed up on their previous study in which they found that male mice sacrifice visual cortex computation precision to save energy in periods of food restriction (Padamsey et al. 2021, *Neuron*). In the present study, the authors find that female mice show much lower levels of adaptation in response to food restriction on the level of metabolic signaling and visual cortex computation. This is an important finding for understanding sex differences in adaptation to food scarcity and also impacts the interpretation of studies employing food restriction in behavioral analyses and learning paradigms.

The manuscript is, in general, very clear and the conclusions are straightforward. The main limitation, that the number of experiments is insufficient to compare the effects of food restriction in males and females directly, is discussed by the authors: to address this point they use Bayes factor analysis to provide an estimate of the likelihood that females and males indeed differ in terms of energy metabolism and sensory processing adaptations during food restriction.

The following points are not entirely clear yet.

1. For a number of experiments the authors use their new data set on females and compare that with the data set previously published on males. In how far are these data sets comparable? Have they been performed originally in parallel for example using siblings of different sexes or have the experiments been conducted several years apart from each other? What is the expected variability, if one repeated these experiments with the same sex

considering the differences/similarities between experimental setups, housing conditions, interindividual differences, etc.?

2. Energy consumption and visual processing may differ between periods in which animals are in different behavioral states. Is there a possibility that male and female mice differed in behavioral state during measurements? Were animals running or resting during visual stimulation and during ATP measurements?

3. Related to the previous point: the authors show that ATP consumption was reduced in male mice during visual stimulation. What about visual cortex ATP consumption in the absence of visual stimulation? Do food-deprived males and/or females show lower ATP consumption in the visual cortex e.g. during sleep?

Reviewer #2 (Public Review):

Summary:

Padamsey et al build up on previous significant work from the same group which demonstrated robust changes in the visual cortex in male mice from long-term (2-3 weeks) food restriction. Here, the authors extend this finding and reveal striking sex-specific differences in the way the brain responds to food restriction. The measures included the whole-body measure of serum leptin levels, and V1-specific measures of activity of key molecular players (AMPK and PPAR α), gene expression patterns, ATP usage in V1, and the sharpness of visual stimulus encoding (orientation tuning). All measures supported the conclusion that the female mouse brain (unlike in males) does not change its energy usage and cortical functional properties on comparable food restriction.

While the effect of food restriction on more peripheral tissue such as muscle and bones has been well studied, this result contributes to our understanding of how the brain responds to food restriction. This result is particularly significant given that the brain consumes a large fraction of the body's energy consumption (20%), with the cortex accounting for half of that amount. The sex-specific differences found here are also relevant for studies using food restriction to investigate cortical function.

Strengths:

The study uses a wide range of approaches mentioned above which converge on the same conclusion, strengthening the core claim of the study.

Weaknesses:

Since the absence of a significant effect does not prove the absence of any changes, the study cannot claim that the female mouse brain does not change in response to food restriction. However, the authors do not make this claim. Instead, they make the well-supported claim that there is a sex-specific difference in the response of V1 to food restriction.

Reviewer #3 (Public Review):

Summary:

The authors food-deprived male and female mice and observed a much stronger reduction of leptin levels, energy consumption in the visual cortex, and visual coding performance in males than females. This indicates a sex-specific strategy for the regulation of the energy budget in the face of low food availability.

Strengths:

This study extends a previous study demonstrating the effect of food deprivation on visual processing in males, by providing a set of clear experimental results, demonstrating the sex-specific difference. It also provides hypotheses about the strategy used by females to reduce energy budget based on the literature.

Weaknesses:

The authors do not provide evidence that females are not impacted by visually guided behaviors contrary to what was shown in males in the previous study.